

The phonetic value of Avestan <ā̊>

Muhammad Rehan

University of California, Los Angeles

rehanmuh@g.ucla.edu

April 24, 2026

Annual Meeting of the American Society for Premodern Asia, Los Angeles

Introduction: Four goals

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3. To argue against the claim of de Vaan (2003) that <ā̊> encodes a retracted vowel [ɑ:].
4. To add support to the claim of Clayton (2020, 90 n. 10) that Avestan <ā̊> most likely represents a rising diphthong [ā̊ə̯].

Roadmap

The data: where do we find <ā̇>?

Proposal: pre-nasal raising

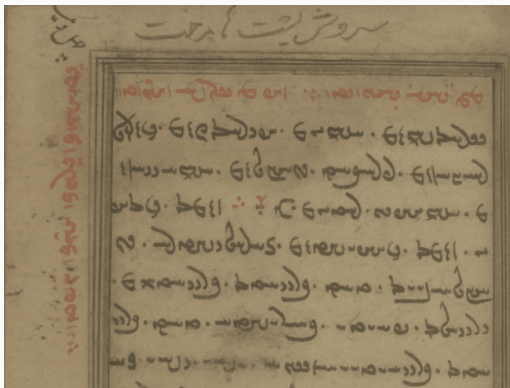
Problems with the retraction/backing hypothesis

Phonetic interpretation

Conclusions

The data: where do we find $\langle \ddot{a} \rangle$?

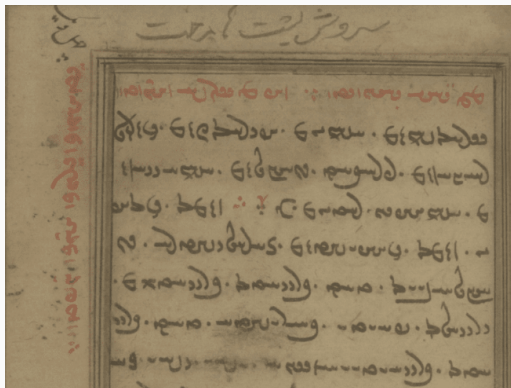
The Avestan manuscripts and script



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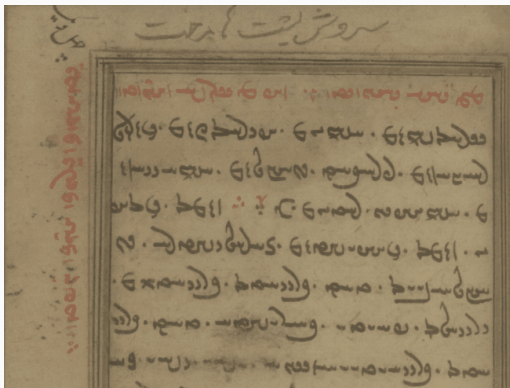
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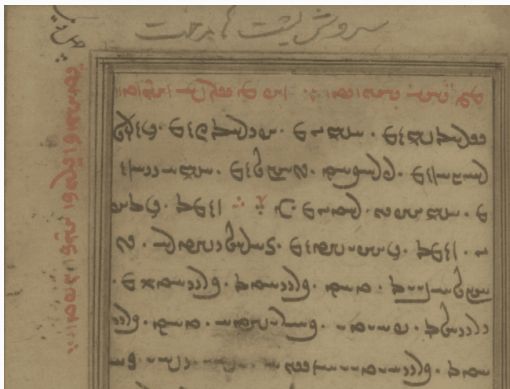
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- Phonetically precise (allophones distinguished) – unlike e.g. Old Persian cuneiform.
- Complication: distinguishing features of the **Sasanian base text** from post-Sasanian scribal practice (Peschl 2022, 71–73).

The Avestan vowel letters: <ā> is a distinct grapheme

𐬀	𐬁	𐬂	𐬃	𐬄	𐬅	𐬆	𐬇	𐬈	𐬉	𐬊	𐬋	𐬌	𐬍	𐬎	𐬏	
a	ā	â	ã	ä	ǣ	ə	ē	e	ē	o	ō	i	ī	u	ū	
[a]	[a:]	[ʊ]	[ʊ:]	[ä]	[ǣ:]	[ə]	[e]	[e:]	[o]	[o:]	[i]	[i:]	[u]	[u:]		
𐬐	𐬑	𐬒	𐬓	𐬔	𐬕	𐬖	𐬗	𐬘	𐬙	𐬚	𐬛	𐬜	𐬝	𐬞	𐬟	
k	x	ḵ	xʷ	g	ḡ	γ	c	j	t	θ/θ	d	ð	ṭ	p	f	b
[k]	[x]	[ḵ]	[xʷ]	[g]	[ḡ]	[γ]	[tʃ]	[dʒ]	[t]	[θ]	[d]	[ð]	[t]	[p]	[f]	[b]
𐬠	𐬡	𐬢	𐬣	𐬤	𐬥	𐬦	𐬧	𐬨	𐬩	𐬪	𐬫	𐬬	𐬭	𐬮	𐬯	𐬰
β	η	ḥ	ηʷ	n	ṇ	ṇ	m	ṁ	ý	y	v	r	s	z	š	
[β]	[η]	[ḥ]	[ηʷ]	[n]	[ṇ]	[ṇ]	[m]	[ṁ]	[j]	[j]	[w]	[r]	[s]	[z]	[ʃ]	
𐬱	𐬲	𐬳	𐬴	𐬵	𐬶	𐬷	𐬸	𐬹	𐬺	𐬻	𐬼	𐬽	𐬾	𐬿	𐬿	𐬿
ž	š	š	h													
[ʒ]	[ʃ]	[ʃ]	[h]													

• The Avestan script contrasts:

- <a> [a] / <ā> [a:]
- <â> (short, marginal; only in MS Pd)
- <ã> (the letter under investigation)
- <ə> [ə] / <ē> [e:]
- <ä> [ä(:)] / <ǣ> [ǣ(:)]

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a	ā	ā̊	ā̋	ə	ē	ē̊	o	ō	i	ī	u	ū			
[a]	[a:]	[ʊ]	[ʊ:]	[ə]	[e]	[e:]	[o]	[o:]	[i]	[i:]	[u]	[u:]			
𐬐	𐬑	𐬒	𐬓	𐬔	𐬕	𐬖	𐬗	𐬘	𐬙	𐬚	𐬛	𐬜	𐬝	𐬞	𐬟
k	x	x̊	x̋	g	g̊	y	c	j	t	θ/θ̊	d	ð	t̥	p	f
[k]	[x]	[x̊]	[x̋]	[g]	[g̊]	[y]	[tʃ]	[dʒ]	[t]	[θ]	[d]	[ð]	[t̥]	[p]	[f]
𐬠	𐬡	𐬢	𐬣	𐬤	𐬥	𐬦	𐬧	𐬨	𐬩	𐬪	𐬫	𐬬	𐬭	𐬮	𐬯
β	ŋ	ŋ̊	ŋ̋	n	n̊	n̋	m	m̊	ẙ	y̋	v	r	s	z	š
[β]	[ŋ]	[ŋ̊]	[ŋ̋]	[n]	[n̊]	[n̋]	[m]	[m̊]	[j̊]	[j̋]	[w]	[r]	[s]	[z]	[ʃ]
𐬰	𐬱	𐬲	𐬳												
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- Hoffmann and Narten (1989, 31) note that the grapheme <ā̊> looks like a ligature of <ā> + <ə>.

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a	ā	ā̊	ā̋	ə	ē	o	ō	i	ī	u	ū					
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 - <ə̊> [ä(:)] / <ə̋> [ä̈(:)]
- Hoffmann and Narten (1989, 31) note that the grapheme <ā̊> *looks like* a ligature of <ā> + <ə>.
- The script has devoted a dedicated letter to this vowel; we need to explain what that vowel was.

Where do we find <â̊>?

Avestan <â̊> is lautgesetzlich in three phonetic contexts:

- (1) a. PIIr. *-ā- *[a:] > Av. -<â̊>- / __ {<ŋc>, <ŋk>, <ŋt>}
- b. PIIr. *-ā- *[a:] > Av. -<â̊>- / __ {<ŋh>, <ŋ̊h>, <ŋ^uh>}
- c. PIIr. *-ā *[a:] > Av. â̊ / __ */h/]_ω

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 - ao (Jackson 1892, xxxviii)

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- [ɑ:] (de Vaan 2003, 383); adopted by Hoffmann and Forssman (1996, 44), Hale (2008, 105), Beguš (2025, 383)

1a. Plr. *ā > YAv. ā̊ / __ <ṇc>, <ṇt>, <ṇk>

YAv. Form	Gloss	Etymology / Cognate
<niiā̊ṇc->	downward-ADV	< PIlr. *[ni-a:nṭj]
<ā̊ṇti>	3.PL.ACT.SBJV ending of thematic verbs	< PIE *[-o-onti]
<f(r)ā̊ṇkauuō>	Mountain-NOM.PL	

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Under Ferrer-Losilla (2016)'s revised interpretation of <ṇ> as a nasal homorganic with the following obstruent, the environment can be phonetically written:

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- (2) a. *-ā̄- *[a:] > YAv. -<ā̄>- / __ [ṇk], [ṇt], [ṇc]

1b PIIr. $^{*}\bar{a}$ - $^{*}[a:]$ > YAv. -< $\overset{\circ}{a}$ >- / __ < ηh >, < $\acute{\eta} h$ >, < $\eta^u h$ >

Avestan	Sanskrit	PIIr. Etym.	Gloss	Translation
<nāṅhābya>	[$'na:sa:b^h ya:m$]	$^{*}['na:sa:b^h ya:m]$	NOSTRIL-DU.INS	'with both nostrils'
< $\overset{\circ}{a}\eta^u harəna$ >-	-	Plr. $^{*}[a:-hwar-ana-]$	PFX-EAT-NMLZ	'for food'
- $\overset{\circ}{a}\acute{\eta} he$	-	-	2SG.MID.SBJV	Verbal ending

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Clayton (2020) has argued that <ŋ> represents a suprasegmental diacritic to mark nasalization on <h> that arose by the process of rhinoglottophilia:

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(3) a. PIIr. *-ā- *[a:] > YAv. -<ā̊>- / __ [h̃], [h̃^j], and [h̃^w]

1c. PIIr. *ā *[a:] > YAv. <ā̊> / __<h>]ω

Avestan	Sanskrit	Etymology (PIIr.)	Translation	Gloss
<buyā̊>	[bʰuːˈyaːs]	*[bʰuːˈyaːs]	‘might you be’	BE-2SG.PRES.ACT.OPT
<haēnayā̊>	[ˈsenaːyːas]	*[ˈsaynaːyːas]	‘of an army’	ARMY-GEN.SG.FEM
<dā̊>	[ˈədʱaːs]	*[ˈadʱaːs]	‘you made’	MAKE.AOR.2SG.ACT.IND

Note: Before enclitics (ca, etc.), becomes -ā̊s

<haēnayā̊sca>	[ˈsenaːyːas=cə]	*[ˈsaynaːyːas=ca]	‘and of the army’	ARMY.GEN.SG=AND
<dā̊scā>	[ˈədʱaːs=cə]	*[ˈadʱaːs=ca]	‘and you made’	MAKE.AOR.2SG=AND

Secondary/analogical attestations of <ā̊>

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The sole productive environment for <ā̊> is *before a tautosyllabic nasal or nasalized consonant*.

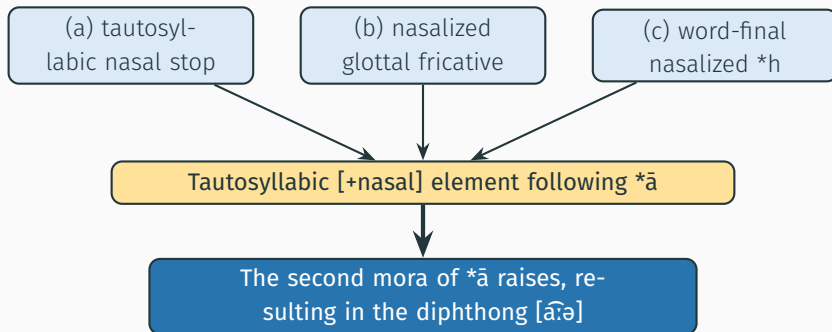
Proposal: pre-nasal raising

Unifying the three environments

- (4) a. Pllr. $*\bar{a}- > \text{Av. } -\langle \overset{\circ}{a} \rangle - / __ [\eta k], [nt], [ɲc]$
b. Pllr. $*\bar{a}- > \text{Av. } -\langle \overset{\circ}{a} \rangle - / __ [\tilde{h}], [\tilde{h}^j], [\tilde{h}^w]$
c. Pllr. $*\bar{a} > \text{Av. } \langle \overset{\circ}{a} \rangle / __ */\tilde{h}/]_{\omega}$

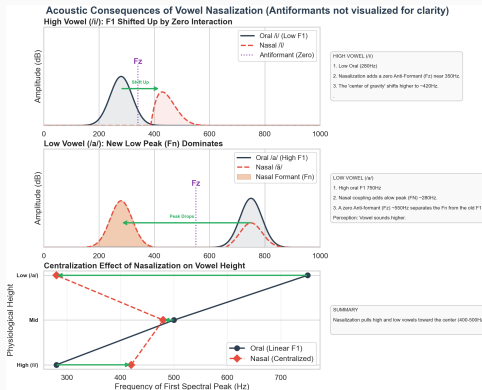
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Why nasalization raises low vowels from an acoustic perspective

Two mechanisms:

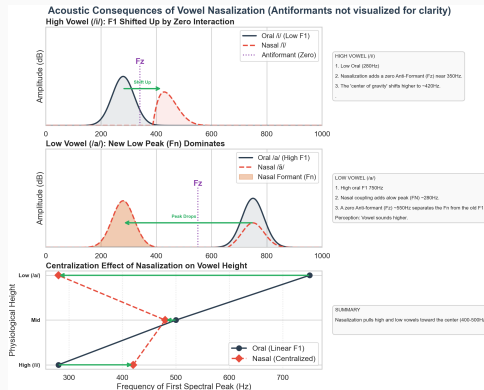


See Ohala (1974, 367) with references; cf. Kawasaki (1986),
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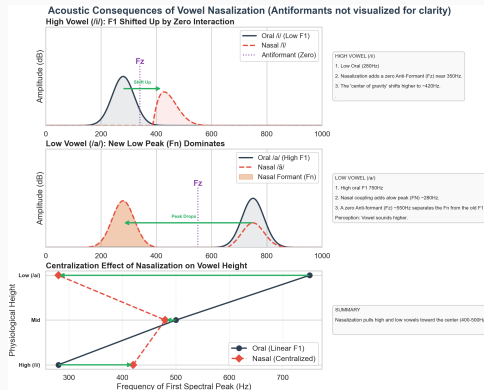


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 - Nasal formant F_{N1} near 250–450 Hz reduces effective F1 of the oral low vowel.

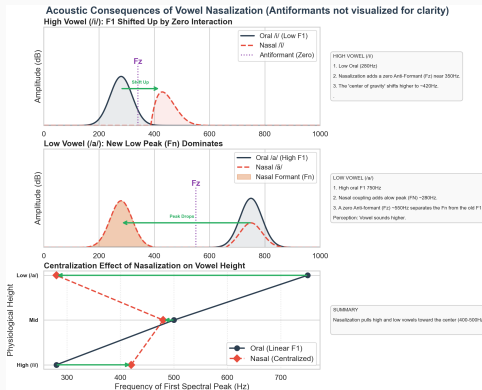


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 - Nasal formant F_{N1} near 250–450 Hz reduces effective F1 of the oral low vowel.
 - Listener reinterprets lowered F1 as a higher vowel (Beddor, Krakow, and Goldstein 1986, 199; Mielke 2008, 333).

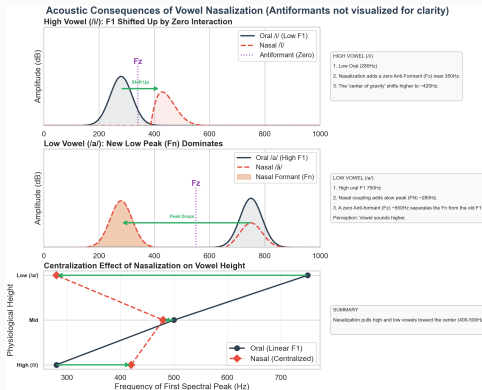


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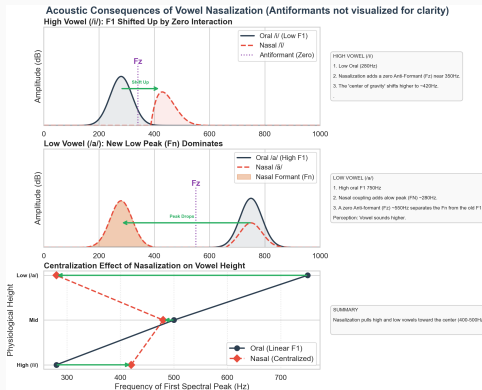


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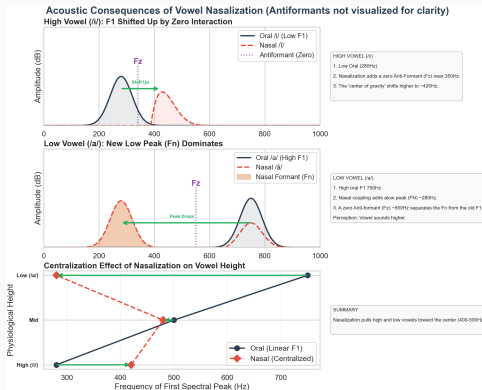
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- Listener *hypercorrects* for expected but weak coarticulation (Beddor, Krakow, and Goldstein 1986, 202).



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Pre-nasal raising is a robust cross-linguistic pattern

It is typologically one of the best-documented vowel shifts. Beddor, Krakow, and Goldstein (1986, 199) note low nasal vowels raise in Breton, Haida, Nama, Seneca, and Zapotec (cf. Hajek 1997).

Language	Input	Output / environment	Reference
Proto-Germanic	*e	*i / __ NC	Lass (1994, 24, 53)
Old English	WGmc *ā	ō / __ N (e.g. <i>mōna</i> ‘moon’)	Lass (1994, 38–42)
Old English	*a	o / __ N (variation <i>mann/monn</i>)	Lass (1994, 38–42)
Romanian	Latin *ā	î / __ N (<i>câmp</i> < <i>campus</i>)	Renwick (2012, 26, 31)
Gascon (dialects)	*ε	e / N__ or __N	Egurtzegi (2018, 174)
Bayonnais Gascon	*a	e / __ N (<i>sent</i> < <i>sanctum</i>)	Egurtzegi (2018)
Zuberoan Basque	*o	u / __ {n, m, ɲ, ħ}	Egurtzegi (2018, 178–9)
American English (NCS)	*æ, *a	raised / __ N (e.g. <i>man</i>)	Dinkin (2011)
Philadelphia English	*aw	raised / __ N	Fruehwald (2012, 30)

Problems with the retraction/backing hypothesis

Problems with the retraction hypothesis before <ηc>, <ηk>, <ηt>

Feature Claim	Retraction Theory	Homorganic Interpretation
	<η> represents a uvular nasal [N] that triggered backing.	<η> represents a nasal homorganic with the consonant it precedes.
Phonetic Realization Implication	Invariably [N] Backing phonetically motivated by the uvular nasal	<ηk> = [ŋk], <ηt> = [nt], <ηc> = [ŋc] Retraction is phonetically unmotivated in <ηt> and <ηc> sequences.
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Context	Phonetic Reality	Phonetic motivation for *[a] > [a:]?
<ŋh>, <ŋ̊h>, <ŋ ^u h>	Articulatory: [<i>h̃</i>], [<i>h̃ʲ</i>], and [<i>h̃^w</i>] (Rhinoglottophilia; Clayton 2020)	No: /h/ lacks [+back] feature; no articulatory motivation for retraction.
	Acoustic: [<i>h̃</i>] (Beguš 2025)	Maybe: Breathiness lowers F2, creating <i>percept</i> of backing.
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The lack of symmetry in word-final development of *-as

Problem	Detail
Retraction	Putative Plr. <i>*ā</i> retraction /_ #h contradicts Plr. <i>*-ah</i> > OAv. <i>ā</i> [ə(:)] (contra Beguš 2025, 9–11)
Vowel Quality	Asymmetry in identical environment (/_ #h): Plr. <i>*/ǎ/</i> raises , whereas <i>*/ā/</i> backs .
Chronology	Changes likely contemporaneous (de Vaan 2003, 384); divergence of <i>*-āh</i> > <ā̊> vs. <i>*-ah</i> > <ā> difficult to explain.

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The <ā> ~<ā̊> alternation pattern is strange for a backing claim

Prediction under [ɑ:] hypothesis: confusion with <ō> [o:] (acoustically adjacent).

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If <ā̊> were [ɑ:], we would expect scribes to occasionally write <ō> (acoustically similar). That they do not is evidence against backing.

Non-raising before high vowels in following syllables

When *ā is followed by /h/ + {i/u(:), y, w}, we find *preserved* <ā>, not <ā̊>:

- (8) a. *māhiia-* (not *mā̊hiia-)
b. *uruuarāhu* (not *uruuarā̊hu)
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Under pre-nasal raising: directly explained by **active velum closure** for high-vowel articulation (Lubker 1968; Ohala 1975). Following high vowels inhibit nasalization spread, so no raising trigger.

Phonetic interpretation

What was the phonetic value?

1. Rising diphthong [ā̃:ə] (Clayton 2020).
 - The ligature *is* the diphthong: <ā̃>+<ə>.

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- Explains why the letter has *three* strokes (ā̃-component) rather than two (ā-component): its first target is long.

Conclusions

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4. The phonetic value is most likely a rising diphthong [ǎ:ə̃].

Thank you!

I am indebted to the members of the PIES Graduate Seminar and the UCLA Phonology Seminar for their help and support. Special thanks to John Clayton and Stephanie Jamison for saving me from numerous errors.

Comments and correspondence welcome: rehanmuh@g.ucla.edu

Appendix

Model A (branching)



Model B (linear)



Figure 1: Two views of the Avestan dialectology (de Vaan 2003, 8); the pre-nasal raising in question predates both.

Latin **ā* > Romanian /*ɨ*/ before N (Renwick 2012, 26, 31):

- (9) a. Latin *campus* ‘field’ > Romanian *câmp* [kɨmp]
b. Latin *lāna* ‘wool’ > Romanian *lână* [lɨnə]
c. Latin *māne* ‘tomorrow’ > Romanian *mâine* [mɨjne]
d. Latin *ambulāre* > Romanian *umbla* [umbla] (additional change)

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- The Romanian reflex /ɨ/ is a close central unrounded vowel — a natural output of a raising trajectory from /a/.
- The environment is precisely the one proposed for Avestan: a *tautosyllabic nasal in the coda*.
- The outcome is *central/high*, not *back*.

Pre-/post-nasal raising of /ɛ/ to /e/ in Gascon, Lengadocian, Provençal (Egurtzegi 2018, 174):

Gascon	Provençal	Latin	Gloss
['bɛŋgo]	['vɛŋgo]	<i>veniō</i>	I come
[ben] / [be] / [bẽ]	[veŋ]	<i>venit</i>	he comes
[bent] / [ben] / [beŋ]	[vẽŋ]	<i>ventum</i>	wind

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Bayonnais Gascon: raising of /a/ before N

Bayonnais Gascon	(Late) Latin	Gloss
[enz]	<i>antius</i> [antius]	but
[ken]	<i>quantum</i> [kʷantũ]	how much
[ten]	<i>tantum</i> [tantũ]	so much
[sen]	<i>sanguem</i> [saŋgʷẽ]	blood
[sent]	<i>sanctum</i> [saŋktũ]	saint

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The parallel is striking: *a tautosyllabic nasal in the coda raises the low vowel to a mid-front vowel* — the exact pattern I propose for Avestan.

Egurtzegi (2018, 178–179): /o/ raises to [u] contiguous to /n, m, ɲ, ñ/, including rhinoglottophilic /ñ/ — exactly the environment I am proposing for Avestan.

Standard Basque	Zuberoan	Transcription	Gloss
<i>ohore</i>	<i>uhure</i>	[u'ħure]	honor
<i>ohoin</i>	<i>uhuĩn</i>	[u'ħuɣɲ]	thief
<i>inor</i>	<i>ihur</i>	[i'ħur]	nobody
<i>inon</i>	<i>ihun</i>	[i'ħun]	nowhere
<i>inoiz</i>	<i>ihuiz</i>	[i'ħuiɣ]	never
<i>laino</i>	<i>lanhú</i>	[lan'ħu]	cloud
<i>anoa</i>	<i>anhua</i>	[an'ħua]	food portion

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The Zuberoan Basque pattern is a precise typological match for the Avestan rhinoglottophilic case (rule 1b above): *V > raised V / __[ħ].

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